NCEA Level 1 Mathematics and Statistics
91947 · Demonstrate mathematical reasoning
Excellence-level practice questions

PRACTICE ANSWER SCHEDULE

About this answer schedule

- This schedule gives the **final answers only** to the 91947 practice excellence questions.
- Solutions are **NOT** provided in this answer schedule to encourage students NOT to just copy off the solution and think they learnt something.
- It is not an official NZQA assessment schedule, and no marks or grades are given. However, solving these levels of questions in the exam should award you an excellence mark.

How to use it

- Only check an answer once you've made a proper attempt at the question.
- If your answer doesn't match, go back and find your mistake before looking at anything else.
- For "show that" questions, the result is given, so it's your working that counts. If you have proved a statement, then your statement counts as the answer, and counts as a solution.

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- If you spot an error or typo, please let me know so I can fix it.

Good luck!

91947 Practice Excellence Questions

Answer Schedule

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Q	Answer
1	14 truck loads (volume = $84\pi \approx 263.9$ m^3; mass ≈ 200.6 tonnes; $200.6 \div 15 \approx 13.4$)
2	(i) $x = 8$, $y = 4$ (each side 20 cm) (ii) $100\sqrt{3} \approx 173$ cm^2
3	(i) $y = 3x^2 + 2$ (ii) $x = 9$
4	(i) AG = 17 cm (AC = 15 cm) (ii) $\approx 28.1°$
5	$\frac{1280\pi}{3} \approx 1340$ cm^3 (radius of the water surface = 8 cm)
6	$\frac{22}{3} \le x < 13$, so $x = 8, 9, 10, 11, 12$
7	(i) $(n-1) + n + (n+1) = 3n$, which is a multiple of 3 (ii) $n + (n+1) + (n+2) + (n+3) = 4n + 6 = 4(n+1) + 2$, which leaves a remainder of 2 on division by 4
8	(i) 15 litres per minute; this is the gradient of the line (ii) $V = 15t + 5$; 5 litres (iii) 33 minutes
9	(i) 5 m by 8 m (from $w^2 + 3w - 40 = 0$) (ii) 30 m^2 (70 – 40)
10	PB ≈ 10.0 cm (AP = $8 \tan 32° \approx 5.00$ cm)
11	$115\pi \approx 361$ cm^2 (cone 65π, hemisphere 50π)
12	(i) Coffee \$6.80, muffin \$2.50 (ii) $c = 2a - b$
13	(i) $y = 2 \times 3^x$ (ii) $x = 7$ (iii) Any three of: y-intercept at $(0, 2)$; always increasing; never negative; horizontal asymptote $y = 0$ as $x \to -\infty$; constant multiplier (ratio) of 3
14	(i) ≈ 31.3 m (ii) ≈ 67.0 m (iii) ≈ 78.0 m
15	30 m by 60 m; greatest area = 1800 m^2
16	(i) 18 cm (length scale factor $\frac{3}{2}$) (ii) 4 : 9
17	(i) $2x^4y^5$ (ii) $x = 3$
18	(i) $y = x^2 - 4x - 12$ (ii) $(2, -16)$ (iii) $(0, -12)$ and $(6, 0)$
19	(i) Shown: $3\pi r^3 + \frac{2}{3}\pi r^3 = \frac{11}{3}\pi r^3$ (ii) $r \approx 5.58$ cm
20	(i) $x = 8$ (sides 5, 12, 13 cm) (ii) 30 cm^2
21	(i) Shown: each of the n isosceles triangles splits into two right-angled triangles with half-base $p \sin\left(\frac{180°}{n}\right)$ and height $p \cos\left(\frac{180°}{n}\right)$ (ii) $\approx 40\,700$ cm^2 ($40\,729$ cm^2)
22	(i) $k = 6$ or $k = -6$ (ii) $-6 < k < 6$
23	(i) $a\sqrt{7}$ (ii) $\approx 61.9°$
24	350 cm^3 (whole pyramid 400 cm^3, small pyramid 50 cm^3)

Q	Answer
25	$(2n+1)^2-(2n-1)^2=8n$, which is a multiple of 8
26	(i) P$(3,-2)$ (ii) 21 units^2 (y-intercepts $(0,-4)$ and $(0,10)$)
27	(i) Shown: length scale factor $\frac{1}{3}$, so volume scale factor $\left(\frac{1}{3}\right)^3=\frac{1}{27}$ (ii) 26 : 1
28	(i) 24 km h^{-1}, then 0 km h^{-1} (at rest), then 45 km h^{-1}; each speed is the gradient of that part of the graph (ii) 27.6 km h^{-1}
29	(i) $T=n^2+3n=n(n+3)$ (ii) Pattern 20
30	(i) $\sqrt{61}\approx 7.81$ m (ii) $\approx 22.6°$
31	$7\le x<10$; $x=7,8,9$
32	(i) Shown: $2\pi r^2+2\pi rh=4\pi r^2\Rightarrow h=r$ (ii) 3 : 4
33	(i) Ladder ≈ 3.51 m; reaches ≈ 3.30 m up the wall (ii) The foot moves ≈ 0.81 m further out; the top slips down ≈ 0.42 m
34	(i) $y=4\times 2^x+3$ (ii) Any three of: y-intercept at $(0,7)$; always increasing; horizontal asymptote $y=3$ as $x\to-\infty$; y is always greater than 3; no x-intercept
35	Exactly $33\frac{1}{3}\%$ (tube volume $2\pi nr^3$, balls $\frac{4}{3}\pi nr^3$, so the balls always fill $\frac{2}{3}$)
36	(i) BC = 15 cm (ii) 9 : 16
37	(i) $k=9$ (ii) $(-1,6)$
38	5 buses and no vans; least cost = \$1250
39	(i) Shown: $\frac{a-s}{s}=\frac{a}{b}\Rightarrow ab=s(a+b)$ (ii) $s=4$ cm
40	3 cm